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Appendix A Control & Instrumentation Tables

1 INTRODUCTION

This document provides a description of the Owner Enabling Works process control philosophy, which is intended to provide a basis for the Process Control System (PCS) programming. The control of the Enabling Works will be integrated with the existing plant control system and follow current site practices and standards.

2 REFERENCE DOCUMENTS

This document must be read in conjunction with:

- Appendix A - Control & Instrumentation Tables, which provide details of individual instruments, control loops, calculations etcetera.
- The latest revisions of the drawings and documents listed in sections 2.1 and 2.2.

2.1 Client/Project Documentation

- Owner DCS Functional Specification, SPEC-00-E-30.
- Owner Enabling Works Instrument List, document 2000-F00-LST-EJ-20001.
- Owner Enabling Works Piping and Instrumentation Diagrams (P&IDs) listed below.

Table 2.1 Piping & Instrumentation Diagrams

Number	Name
2000-F75-PID-PR-25001	Water P&ID – Scheme and Fire Water Pumps
2000-F75-PID-PR-25002	Water P&ID – Potable Water
2000-F75-PID-PR-25003	Fire Water P&ID
2000-F75-PID-PR-25004	Fire Water P&ID
2000-F75-PID-PR-25005	Fire Water P&ID
2000-F75-PID-PR-25006	Fire Water P&ID

2.2 Vendor Documents

The following table will be populated on receipt of certified vendor data.

Table 2.2 Vendor Documents

Number	Name
Fire Water Pumps – Dynapumps	
2000-ME006-E02-25001	Fire Water Pump Skid PID
Potable Water Pumps – Dynapumps	
2000-ME006-E02-25002	Potable Water Pump Skid PID

3 NOMENCLATURE

This document may use slightly different nomenclature to that currently in use at Owner Given the Enabling Works are to be integrated into the existing plant and control system, the nomenclature currently in use will take precedence. The following sections provide explanation for the terms used within this document and appendices.

3.1 Drive Interlocks

There are three types of interlocks used in this control philosophy, as described below:

3.1.1 Critical Interlocks

These interlocks prevent damage to plant equipment or danger to personnel. They exclude emergency stops and protection relay alarms which are monitored and interlocked separately as part of the standard drive logic. Critical interlocks trip a drive in both remote and local modes.

3.1.2 Process Interlocks

Process interlocks are used to stop drives to prevent process spillage or poor plant performance. Process interlocks trip a drive in remote mode only.

3.1.3 Start Permissive

Start permissives are interlocks that prevent equipment from starting but do not stop the drive once it is running.

3.1.4 Interlock Bypass

Bypassing of interlocks will be undertaken in accordance with current Owner practice and procedures.

3.2 Valve Interlocks

Actuated (on/off) valves will have interlocks similar to those for drives, as described below.

Table 3.1 Valve Interlocks

Interlock	Description
Open Permissive	Condition must be true for valve to be permitted to open
Close Permissive	Condition must be true for valve to be permitted to close
Open Trip	Valve will open on given condition
Close Trip	Valve will close on given condition

4 PROCESS CONTROL BY AREA

The following section describes the operation, control logic, and a summary of the interlocks for the Enabling Works.

Details of instrument settings, controllers, interlocks, alarms, and other control functions are contained in the control philosophy tables located in Appendix A.

4.1 Water Services

The upgraded water services area consists of:

- Two new Scheme and Fire Water Tanks.
- One new standalone fire water pump skid, consisting of:
  - Electric jockey pump.
  - Duty electric fire water pump.
  - Diesel backup fire water pump.
- Two duty electric Scheme water pumps.
- One diesel backup Scheme water pump.
- A new Potable Water Tank.
- New standalone potable water pump skid consisting of:
  - Duty electric potable water pump.
  - Diesel backup potable water pump.

The following P&IDs should be referred to in conjunction with the description of the water services area control philosophy:

- 2000-F75-PID-PR-25001.
- 2000-F75-PID-PR-25002.
- 2000-F75-PID-PR-25003.
- 2000-F75-PID-PR-25004.
- 2000-F75-PID-PR-25005.
- 2000-F75-PID-PR-25006.

4.1.1 2000 Process Description

Scheme water from the existing supply source will be stored in two new tanks (“Scheme and Fire Water Tanks”). The tanks have been sized to provide sufficient scheme water capacity for the plant, as well as a reserve of fire water. To maintain the fire water reserve, the Scheme Water Pumps draw from a take-off point higher on the tank walls than the Fire Water Pumps. Both the Scheme and Fire Water Pumps have suction piping that draws from both tanks.

Duty/duty/standby Scheme Water Pumps will draw scheme water from the Scheme and Fire Water Tanks and reticulate scheme water to the existing scheme water system. The pumps are sized such that they can meet the scheme water demand with two pumps in operation. The scheme water system pressure will be maintained to a set value by a mechanical pressure sustaining valve in a return line to the scheme and fire water tanks.

The fire water system pressure will be maintained by a jockey pump. When a large decrease in fire water system pressure occurs (i.e., when a hydrant is opened), the duty fire water pump will start automatically. The backup fire water diesel pump will start automatically if the pressure in the system continues to fall. This pump will continue to run until manually stopped.

Potable water will be boosted to the required pressure by a dedicated potable water pump package and reticulated to the existing potable water system. The potable water system pressure will be maintained

by an electric pump. The backup diesel pump will start automatically if the pressure in the system falls below a given point or if power is not available to start the duty pump.

4.1.2 Scheme and Fire Water Tanks

Each scheme and fire water tank will be fitted with a level element to provide monitoring and alarming. **Both F75-LIT-1001A and F75-LIT-1001B will be inputs to F75-LI-1001 though only one of the values, selected from the DCS, will be displayed and used for level control of the tanks.** The level in the tanks is maintained by cycling the on/off valve supplying the tank to maintain water level in the tanks. ***(1)***

4.1.3 Scheme Water Pumps

The fixed speed scheme water pumps can be started and stopped from either the OIS or the LCS. A pressure sustaining valve fitted on the recirculation line back to the scheme and fire water tanks will prevent dead heading of the pumps and to prevent over pressurisation of the scheme water system on low water demand. The standby diesel scheme water pump will start if the pressure in the system falls below a given point, pump will stop when pressure in the system reaches set point.

4.1.4 Fire Water Pump Skids

The fire water pumps will be controlled through a standalone vendor panel on the skid. The jockey pump will start and stop as required to maintain the overall system pressure. The electric fire water pump will start on detection of low pressure in the fire water system. Should the pressure fall further, the diesel pump will automatically start. The local control panels will provide status information for each pump to the PCS.

4.1.5 Potable Water Tank

Potable water from the existing supply will be stored in the potable water tank. The tank will be fitted with a level element to provide monitoring and maintain water level in the tank. The level in the tank is maintained by cycling the on/off valve supplying the tank to maintain water level in the tanks. The low-low level in the potable water tank will be used by the potable water pump skid LCPs for interlocking.

4.1.6 Potable Water Pump Skid

The potable water pump skid will include a duty electric pump and backup diesel pump that will be controlled by dedicated Local Control Panels mounted on the skid. The duty pump will **run continuously, with a minimum flow maintained by a pressure sustaining valve on the return line to the potable water tank.** ~~start on detection of low pressure in the potable water system.~~ The backup diesel pump will start automatically if the pressure in the system continues to fall. The backup diesel pump will also start automatically should power not be available to start the duty pump. Pump will start when pressure falls below a given point**, set at 500 kPa during commissioning** and stop when pressure in the system reaches the given point **(set at 925 kPa)**. The local control panels will provide status information for each pump to the PCS. ***(1)***

A pressure ~~sustaining~~ **relief** valve fitted to the discharge line of each pump will ~~allow a minimal flow through~~ **provide dead-head protection for** the pumps~~ to be maintained during periods of low water demand to prevent overheating of the pumps.~~ *(1)*

Appendix A Control & Instrumentation Tables

Table of Contents		Total	On Hold	Complete (%)	FAT Complete (%)	SAT Complete (%)
1	Analogue	7	0	100.0	0.0	0.0
2	Digital	19	0	100.0	0.0	0.0
3	Totalisers	2	0	100.0	0.0	0.0
4	Open/Close Valves	8	0	100.0	0.0	0.0
5	PID Controllers	0	0	0.0	0.0	0.0
6	Miscellaneous Controllers	0	0	0.0	0.0	0.0
7	Interlocks	17	0	100.0	0.0	0.0
8	Sequences (Steps)	14	0	100.0	0.0	0.0
9	Calculations	0	0	0.0	0.0	0.0
	Total	67	0	100.0	0.0	0.0

Alarms	Priority	Count	%
1	Low	20	42.6
2	Med	4	8.5
3	High	14	29.8
4	Critical	9	19.1
	Total	47	100.0

Abbreviations	
FC	Fail closed
FO	Fail open
FL	Fail last
LCP	Local control panel
LCS	Local control station
A_L	Low alarm
A_H	High alarm
A_LL	Low-Low alarm
A_HH	High-High alarm
P_L	Low process switchpoint
P_H	High process switchpoint
P_LL	Low-Low process switchpoint
P_HH	High-High process switchpoint
OIT	Operator interface terminal
Range	Instrument range
SEQ	Sequence
TBA	To be advised
v	Vendor controlled
VTA	Vendor to advise
x	PCS controlled
*	Denotes a sequence stop in place of an interlock

ANALOGUE INSTRUMENTS

Tag Source	P&ID	Description	Supplier	Hold	Range			Process Switchpoints				Alarms				Alarm Priority				Delay (s)	Alarm Masking	Controller	Comments	Rev
					Min	Max	Unit	P_LL	P_L	P_H	P_HH	A_LL	A_L	A_H	A_HH	A_LL	A_L	A_H	A_HH					
F75-LT-1001A	2000-F75-PID-PR-25001	Scheme and Fire Water Tank 1 Level Transmitter	1010		0	100	%	-	80	98	-	10	54	70	99	Critical	Critical	High	Low	5		-	10% level is above fire water pump suction 54% level is above scheme water pump suction	1
F75-LT-1001B	2000-F75-PID-PR-25001	Scheme and Fire Water Tank 2 Level Transmitter	1010		0	100	%	-	80	98	-	10	54	70	99	Critical	Critical	High	Low	5		-	10% level is above fire water pump suction 54% level is above scheme water pump suction	1
F75-LT-1002	2000-F75-PID-PR-25002	Potable Water Tank Level Transmitter	1010		0	100	%	-	80	98	-	5	70	99	-	High	High	Low	-	5		-		0
F75-PT-1003	2000-F75-PID-PR-25002	Potable Water Pumps Discharge Pressure Transmitter	1010		0	1,000	kPa	-	-	-	-	-	200	-	750	-	Med	-	High	5		-		0
F75-PT-1024	2000-F75-PID-PR-25002	Potable Water Distribution Pressure Transmitter	1010		0	1,000	kPa	-	-	-	-	-	200	-	750	-	Med	-	High	5		-		0
F75-PT-1035	2000-F75-PID-PR-25002	Diesel Potable Water Pump Discharge Pressure Transmitter	M1		0	1,100	kPa	-	500	925	-	-	-	-	-	-	-	-	-	5		-	Vendor controlled	1
F75-PT-1043	2000-F75-PID-PR-25001	Diesel Scheme Water Pump Discharge Pressure Transmitter	1010		0	1,100	kPa	-	500	690	-	-	480	710	-	-	-	-	-	5		-	TBC during commissioning	0

DIGITAL INSTRUMENTS

Tag Source	P&ID	Description	Hold	Alarm Priority	Alarm Masking	Comments	Rev
F75-PW-0488RUN	2000-F75-PID-PR-25001	Electric Fire Water Pump Running		Critical			0
F75-PW-0488PWR	2000-F75-PID-PR-25001	Electric Fire Water Pump Power Fail		High			0
F75-PW-0488FLT	2000-F75-PID-PR-25001	Electric Fire Water Pump Charger Not Healthy		High			0
F75-PW-0486ALM	2000-F75-PID-PR-25001	Diesel Fire Water Pump Common Fault Alarm		Critical			0
F75-PW-0486RUN	2000-F75-PID-PR-25001	Diesel Fire Water Pump Running		Critical			0
F75-XI-1013	2000-F75-PID-PR-25001	Diesel Fire Water Pump Diesel Tank Low Level Alarm		Critical			0
F75-XI-1014	2000-F75-PID-PR-25001	Diesel Fire Water Pump Diesel Tank Leakage Alarm		High			0
F75-PW-0487RUN	2000-F75-PID-PR-25001	Fire Water Jockey Pump Running		-			0
F75-PW-0487FLT	2000-F75-PID-PR-25001	Fire Water Jockey Pump Not Healthy		-			0
F75-PW-0484RUN	2000-F75-PID-PR-25002	Electric Potable Water Pump Running		-			0
F75-PW-0484FLT	2000-F75-PID-PR-25002	Electric Potable Water Pump Fail		High			0
F75-PW-0484EN	2000-F75-PID-PR-25002	Electric Potable Water Pump Enable		Low			0
F75-PW-0485RUN	2000-F75-PID-PR-25002	Diesel Potable Water Pump Running		High			0
F75-PW-0485ALM	2000-F75-PID-PR-25002	Diesel Potable Water Pump Common Alarm		Critical			0
F75-PW-0485PWR	2000-F75-PID-PR-25002	Diesel Potable Water Pump Power Supply Not Healthy		High			0
F75-PW-0485EN	2000-F75-PID-PR-25002	Diesel Potable Water Pump Enable		Low			0
F75-PW-0483PWR	2000-F75-PID-PR-25001	Diesel Scheme Water Pump Power Supply Not Healthy		High			0
F75-PW-0483ALM	2000-F75-PID-PR-25001	Diesel Scheme Water Pump Common Fault Alarm		High			0
F75-PW-0483RUN	2000-F75-PID-PR-25001	Diesel Scheme Water Pump Running		-			0
F75-PW-0483EN	2000-F75-PID-PR-25001	Diesel Scheme Water Pump Enable		Low			0
F75-ZSO-1001	2000-F75-PID-PR-25001	Scheme & Fire Water Tank Water Addition Valve Open Position		Low			0
F75-ZSC-1001	2000-F75-PID-PR-25001	Scheme & Fire Water Tank Water Addition Valve Close Position		Low			0
F75-ZSO-1002	2000-F75-PID-PR-25001	Potable Water Tank Water Addition Valve Open Position		Low			0
F75-ZSC-1002	2000-F75-PID-PR-25001	Potable Water Tank Water Addition Valve Close Position		Low			0

DIGITAL INSTRUMENTS

Tag Source	P&ID	Description	Hold	Alarm Priority	Alarm Masking	Comments	Rev
F75-ZSO-1007	2000-F75-PID-PR-25001	Scheme Water Pump 1 Suction Vave Open Position		Low			0
F75-ZSC-1007	2000-F75-PID-PR-25001	Scheme Water Pump 1 Suction Vave Closed Position		Low			0
F85-ZSO-1008	2000-F75-PID-PR-25001	Scheme Water Pump1 Discharge Vave Open Position		Low			0
F85-ZSC-1008	2000-F75-PID-PR-25001	Scheme Water Pump 1 Discharge Vave Closed Position		Low			0
F95-ZSO-1009	2000-F75-PID-PR-25001	Scheme Water Pump 2 Suction Vave Open Position		Low			0
F95-ZSC-1009	2000-F75-PID-PR-25001	Scheme Water Pump 2 Suction Vave Closed Position		Low			0
F85-ZSO-1010	2000-F75-PID-PR-25001	Scheme Water Pump 2 Discharge Vave Open Position		Low			0
F85-ZSC-1010	2000-F75-PID-PR-25001	Scheme Water Pump 2 Discharge Vave Closed Position		Low			0
F85-ZSO-1011	2000-F75-PID-PR-25001	Diesel Scheme Water Pump Suction Vave Open Position		Med		Status feedback on the valve. if does not open within 40 seconds, alarm goes off.	0
F85-ZSC-1011	2000-F75-PID-PR-25001	Diesel Scheme Water Pump Suction Vave Closed Position		Low			0
F85-ZSO-1012	2000-F75-PID-PR-25001	Diesel Scheme Water Pump Discharge Vave Open Position		Med		Status feedback on the valve. if does not open within 40 seconds, alarm goes off.	0
F85-ZSC-1012	2000-F75-PID-PR-25001	Diesel Scheme Water Pump Discharge Vave Closed Position		Low			0

TOTALISERS

						Reset Period				
Tag Source	Type	P&ID	Source Description	Hold	Units	Shift	Day	Cont	Comments	Rev
F75-FQI-1039	FQI	2000-F75-PID-PR-25006	Scheme Water Flow Transmitter (Watercorp)		m³/h	x	x	x	Woltmann-type meter	0
F75-FQI-1040	FQI	2000-F75-PID-PR-25006	Scheme Water Flow Transmitter (Verification)		m³/h	x	x	x	Woltmann-type meter	0

2000-F00-PHL-PR-25001 - APPENDIX A

OPEN-CLOSE VALVES

				Open					Close					Interlocks						
Tag Source	P&ID	Description	Hold	LCS	LCP	OIS	SEQ	Control (CONDITION)	LCS	LCP	OIS	SEQ	Control (CONDITION)	Description	Open Permissive	Close Permissive	Open Trip	Close Trip	Comments	Rev
F75-XV-1001	2000-F75-PID-PR-25001	Scheme and Fire Water Tank Level Control Valve		-	-	x	-	F75-LY-1001 < P_L	-	-	x	-	F75-LY-1001> P_H						ensure the valves open slowly and close slowly to reduce water hammering issue.	0
F75-XV-1007	2000-F75-PID-PR-25001	Scheme Water Pump 1 Suction Valve		-	-	x	-		-	-	x	-								0
F75-XV-1008	2000-F75-PID-PR-25001	Scheme Water Pump 1 Discharge Valve		-	-	x	-		-	-	x	-								0
F75-XV-1009	2000-F75-PID-PR-25001	Scheme Water Pump 2 Suction Valve		-	-	x	-		-	-	x	-								0
F75-XV-1010	2000-F75-PID-PR-25001	Scheme Water Pump 2 Discharge Valve		-	-	x	-		-	-	x	-								0
F75-XV-1011	2000-F75-PID-PR-25001	Scheme Water Pump 3 Suction Valve		-	-	x	-		-	-	x	-								0
F75-XV-1012	2000-F75-PID-PR-25001	Scheme Water Pump 3 Discharge Valve		-	-	x	-		-	-	x	-								0
F75-XV-1002	2000-F75-PID-PR-25002	Potable Water Tank Level Control Valve		-	-	x	-	F75-LIT1002 < P_L	-	-	x	-	F75-LIT1002> P_H						ensure the valves open slowly and close slowly to reduce water hammering issue.	0

DRIVE INTERLOCKS AND CONTROL

					Start					Stop (Non-Latching)					Interlocks						
Tag Source	P&ID	Name	Drive Type	Hold	LCS	LCP	OIS	SEQ	Condition	LCS	LCP	OIS	SEQ	Condition	Description	Start	Process	Critical	Delay (s)	Comments	Rev
75-PW-481	2000-F75-PID-PR-25001	SCHEME WATER PUMP 1	415 DOL		x	-	x	x		x	-	x	x								0
															F75-ZSO-1007 NOT ACTIVE		x		10	Suction Valve	0
															F75-ZSC-1008 ACTIVE		x		10	Discharge Valve	1
															F75-LI-1001 < A_L		x			Tank Level Low	1
75-PW-482	2000-F75-PID-PR-25001	SCHEME WATER PUMP 2	415 DOL		x	-	x	x		x	-	x	x								0
															F75-ZSO-1009 NOT ACTIVE		x		10	Suction Valve	0
															F75-ZSC-1010 ACTIVE		x		10	Discharge Valve	1
															F75-LI-1001 < A_L		x			Tank Level Low	1
75-PW-483	2000-F75-PID-PR-25001	DIESEL SCHEME WATER PUMP 3	N/A		x	-	x	-		x	-	x	-								1
															F75-ZSO-1011 NOT ACTIVE		x		10	Suction Valve	0
															F75-ZSO-1012 NOT ACTIVE		x		10	Discharge Valve	0
															F75-LI-1001 < A_L		x			Tank Level Low	1
75-PW-486	2000-F75-PID-PR-25001	DIESEL FIRE WATER PUMP	N/A		-	v	-	-		-	v	-	-		VENDOR CONTROLLED						0
75-PW-487	2000-F75-PID-PR-25001	FIRE WATER JOCKEY PUMP	415 DOL		-	v	-	-		-	v	-	-		VENDOR CONTROLLED						0
75-PW-488	2000-F75-PID-PR-25001	FIRE WATER PUMP	415 DOL		-	v	-	-		-	v	-	-		VENDOR CONTROLLED						0
75-PW-484	2000-F75-PID-PR-25002	POTABLE WATER PUMP	415 DOL		-	v	-	-		-	v	-	-		**F75-LT-1002 < A_LL**		**x**				**1**
75-PW-485	2000-F75-PID-PR-25002	DIESEL POTABLE WATER PUMP	N/A		-	v	-	-		-	v	-	-		**F75-LT-1002 < A_LL**		**x**				**1**

SEQUENCES

Seq #	Sequence Description	Hold	Step	Action	Tag / Item	Description	Delay After Step (s)	Step Advance Condition	Comments	Rev
F75-SEQ-001	Scheme Water Pump 1 Start Sequence		1	CONFIRM	F75-LT-1001	Scheme & Fire Water Tank Level		F75-LT-1001.PV > LL		0
			2	OPEN	F75 -XV-1007	Scheme Water Pump 1 Suction Valve	40	F75-ZSO-1007 ACTIVE	Valve open/close time minimum 35s	1
			3	OPEN	F75 -XV-1008	Scheme Water Pump 1 Discharge Valve	10	F75-ZSC-1008 NOT ACTIVE	Valve open/close time minimum 35s	1
			4	START	F75 -PW-481	Scheme Water Pump 1 START		F75-PW-481 RUNNING		0
F75-SEQ-002	Scheme Water Pump 1 Stop Sequence		1	STOP	F75 -PW-481	Scheme Water Pump 1 STOP		F75-PW-481 STOPPED		0
			2	CLOSE	F75 -XV-1008	Scheme Water Pump 1 Discharge Valve	40	F75-ZSC-1008 ACTIVE	Valve open/close time minimum 35s	1
			3	CLOSE	F75 -XV-1007	Scheme Water Pump 1 Suction Valve	40	F75-ZSC-1007 ACTIVE	Valve open/close time minimum 35s	1
F75-SEQ-003	Scheme Water Pump 2 Start Sequence		1	CONFIRM	F75-LT-1001	Scheme & Fire Water Tank Level		F75-LT-1001.PV > LL		0
			2	OPEN	F75 -XV-1009	Scheme Water Pump 2 Suction Valve	40	F75-ZSO-1009 ACTIVE	Valve open/close time minimum 35s	1
			3	OPEN	F75 -XV-1010	Scheme Water Pump 2 Discharge Valve	10	F75-ZSC-1010 NOT ACTIVE	Valve open/close time minimum 35s	1
			4	START	F75 -PW-482	Scheme Water Pump 2 START		F75-PW-482 RUNNING		0
F75-SEQ-004	Scheme Water Pump 2 Stop Sequence		1	STOP	F75 -PW-482	Scheme Water Pump 2 STOP		F75-PW-482 STOPPED		0
			2	CLOSE	F75 -XV-1010	Scheme Water Pump 2 Discharge Valve	40	F75-ZSC-1010 ACTIVE	Valve open/close time minimum 35s	1
			3	CLOSE	F75 -XV-1009	Scheme Water Pump 2 Suction Valve	40	F75-ZSC-1009 ACTIVE	Valve open/close time minimum 35s	1